

Investigate how advances in technologies enable advances in science, and how science has contributed to developments in technologies and engineering

Learning intention

- investigate how advances in technologies enable advances in science, and how science has contributed to developments in technologies and engineering.

Teach and model

- investigating how satellites generate global data including ocean temperatures.
- recognising that the development of fast computers has made possible the analysis of DNA sequencing.
- exploring how the development of new materials and thin films has led to better computer chips and solar cells.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.